

Ex:No: 2

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Data Preprocessing and Integration of Customer Purchase Data from Excel and Relational Database

Problem Description:

A retail organization maintains customer purchase information for business analytics. The data is stored across **two different sources**:

- **Online purchases** → Excel workbook
- **In-store purchases** → Relational database (SQLite)

Each dataset contains the following attributes:

- **CustomerID** – Unique identifier for each customer
- **Age** – Numerical attribute representing customer age
- **PurchaseAmount** – Numerical attribute representing purchase value
- **Membership** – Categorical attribute indicating membership type (Gold, Silver, Regular)
- **City** – Categorical attribute representing customer location

Due to manual data entry and synchronization issues, the datasets contain **missing values**, **duplicate records**, and **inconsistent formats**.

The organization needs to load both datasets into R, integrate them, explore the data, separate numerical and categorical attributes, clean the data, and prepare a consistent dataset for analysis.

Objective:

To load customer purchase data from an **Excel workbook** and a **relational database** into R, integrate the datasets, explore the data, separate numerical and categorical attributes, handle missing values and duplicates, convert categorical variables into factors, and prepare a clean dataset for business analysis.

Data Preprocessing

Data preprocessing in R is a crucial step to prepare your data for analysis or machine learning tasks. Below are some common steps along with example code snippets.

Steps in Data Preprocessing

1. Data Collection:

- Load data from various sources such as CSV files, Excel files, databases, etc.

2. Data Cleaning:

- Handle missing values. (In R, the `na.rm` parameter is used to exclude missing values (NA) during calculations. This ensures accurate results by ignoring NA values.)
- Remove duplicates.
- Correct errors and inconsistencies.

3. Data Transformation:

- Normalize or standardize data.
- Encode categorical variables.
- Apply feature scaling techniques.

4. Data Integration:

- Combine data from multiple sources.
- Resolve schema and data conflicts.

5. Data Reduction:

- Reduce the size of the dataset while retaining important information.
- Use techniques like Principal Component Analysis (PCA) or feature selection.

6. Data Splitting:

- Split the data into training and testing sets.

Data Preprocessing Tasks

- Data collection from relational database
- Data integration from multiple tables
- Handling missing numerical and categorical values
- Removing duplicate records
- Converting categorical variables into factors
- Preparing a clean dataset for analysis

Procedure:

• Step 1: Load Required Packages

install.packages("Openxlsx")

install.packages("DBI")

install.packages("RSQLite")

library(Openxlsx)

library(DBI)

library(RSQLite)

Customer Id	Age	purchase Amount	Membership	city
1	25	1200	Gold	Chennai
2	30	1500	Silver	Bangalore
3	<NA>	1800	Gold	Chennai
4	45	<NA>	<NA>	Mumbai
5	40	2000	Silver	<NA>

• Step 2: Create and Store Excel Dataset (Online Purchases)

	CustomerID	Age	PurchaseAmount	Membership	City
1	1	25	1200	Gold	Chennai
2	2	30	1500	silver	Bangalore
3	3	NA	1800	Gold	Chennai
4	4	45	NA	<NA>	Mumbai
5	5	40	2000	silver	<NA>

Online_data = data.frame(customerId = c(1,2,3,4,5)

Age = c(25,30,NA,45,40),

Purchase Amount = c(1200,1500,1800,NA,2000),

membership = c("Gold", "Silver", "Gold", NA, "silver"),

City = c("Chennai", "Bangalore", "Chennai", "Mumbai",
NA))

#createWorkbook wb

wb = createWorkbook()

addWorksheet(wb, "Online Purchases")

writeData(wb, "OnlinePurchases", Online_data)

SaveWorkbook(wb, "online-purchases.xlsx", Overwrite = TRUE)

Print output online_data

print(Online_data)

> Customer data

customer.Id	Age	Purchase Amount	membership	city
1	25	1200	Gold	chennai
2	30	1300	Silver	NA
3	35	1500	NA	Bangalore
4	NA	1000	Regular	mumbai
5	45	2000	NA	Kolkata
6	50	1800	Gold	NA
7	20	2200	Gold	chennai
7	NA	NA	Silver	mumbai

> str(customer_data)

data frame: 10 Obs. of 5 variables

```
$ customer.Id : num 12 3 4 5 6 7 7
$ Age : num 25 30 35 NA 45 50 20 NA
$ Purchase Amount : num 1200 1300 1500 1000 2000 2200 NA
$ membership : "Gold" "Silver" "Gold" NA ...
$ city : "chennai", "Mumbai", "Bangalore", NA...
```

- Step 3: Create and Store SQLite Dataset (In-Store Purchases)

	CustomerID	Age	PurchaseAmount	Membership	City
1	3	35	1600	Gold	Chennai
2	4	45	1700	Regular	Mumbai
3	6	28	1400	Silver	Delhi
4	7	NA	1300	Gold	Delhi
5	7	NA	1300	Gold	Delhi

```
con = db Connect (sqlite3, "store_purchases.db")
```

```
instore_data = data.frame (
```

```
  CustomerID = c (3, 4, 6, 7, 7),
```

```
  Age = c (35, 45, 28, NA, NA),
```

```
  PurchaseAmount = c (1600, 1700, 1400, 1300, 1300),
```

```
  membership = c ("Gold", "Regular", "Silver", "Gold", "Gold")
```

```
  , City = c ("Chennai", "Bangalore", "Mumbai", "Delhi", "Delhi")
)
```

```
dbwriteTable (con, "InStorePurchases", instore_data, overwrite = TRUE)
```

```
# Print output instore_data
```

```
Print (instore_data)
```

- Step 4: Load Both Datasets into R

```
excel_df = read_xlsx ("online_purchase.xlsx")
```

```
con = db Connect (sqlite3, "store_purchases.db")
```

```
db_df = dbReadTable (con, "InStorePurchases")
```

```
dbDisconnect (con)
```


- Step 5: Integrate Datasets

```
customer_data = rbind (excel_df, db_df)
```

```
# Print integrated dataset customer_data
```

```
print (customer_data)
```

- Step 6: Data Exploration

```
# Structure customer_data. Use str()
```

```
str (customer_data)
```

```
# Summary customer_data
```

```
summary (customer_data)
```

```
# Sum of Missing values based on column in customer_data
```

```
colSums (is.na (customer_data))
```

> Summary (Customer_data)

	customer_Id	Age	Purchase Amount	membership	city
min	: 1.00	min: 25.00	min: 1200	length: 10	
1st Qu	: 3.00	1st Qu: 29.00	1st Qu: 1300	class: character	
median	: 4.00	median: 35.00	median: 1500	mode: character	
Mean	: 4.20	mean: 35.43	mean: 1533		
3rd Qu	: 5.75	3rd Qu: 42.50	3rd: 1700		
max	: 7.00	max: 45.00	Max: 2000		
		NA: 3	NA's: 1		

> columns (is.na(customer_data))

customer Id	Age	Purchase Amount	membership	city
0	3	1	1	1

> Sum (duplicated (customer_data))

[1] 1

> table (customer_data \$ membership)

Gold	Regular	Silver
5	1	3

> table (customer_data \$ city)

Bangalore	chennai	Delhi	mumbai
1	3	3	2

Sum of Duplicate records in customer_data

```
sum(duplicate(customer_data))
```

Categorical distribution Membership and City in customer_data. Use table()

```
table(customer_data$membership)
```

```
table(customer_data$city)
```

- Step 7: Separate Numerical and Categorical Data

```
numerical_data = customer_data[, $apply(customer_data, is.numeric)]
```

```
categorical_data = customer_data[, $apply(customer_data, is.character)]
```

```
$apply(customer_data, is.factor)
```

Print numerical data

```
print(numerical_data)
```

Print categorical data

```
print(categorical_data)
```

customer id	Age	Purchase Amount
1	25	1200
2	30	1300
3	NA	1500
4	45	NA
5	40	2000
6	NA	25000
7	NA	3000
7.	28	1000

> categorical data:

membership

city

Gold

chennai

Silver

Bangalore

Regular

mumbai

Silver

chennai

Gold

chennai

Regular

mumbai

Regular

Kolkata

- Step 8: Data Cleaning

Handle missing numerical values in Age and PurchaseAmount

```
numerical_data $ Age [is.na(numerical_data $ Age)] <-
```

```
mean(numerical_data $ Age, na.rm = TRUE)
```

```
numerical_data $ PurchaseAmount [is.na(numerical_data $ PurchaseAmount)] <-
```

```
mean(numerical_data $ PurchaseAmount, na.rm = TRUE)
```

Handle missing categorical values Membership and City

```
categorical_data $ membership [is.na(categorical_data $ membership)] <-
```

```
"Regular"
```

```
categorical_data $ city [is.na(categorical_data $ city)] <- "unknown"
```

Remove duplicates

```
final_data = cbind(numerical_data, categorical_data)
```

```
final_data = final_data [! duplicated(final_data), ]
```

Convert categorical to factor Membership and City

```
final_data $ membership = factor(final_data $ membership)
```

```
final_data $ city = factor(final_data $ city)
```


Customer Id	Age	purchase Amount	membership	city
1.	25.0000	1200.000	Gold	chennai
2.	30.0000	1500.000	Regular	Bangalore
3.	35.42857	1800.000	Silver	mumbai
4.	45.0000	1533.000	Regular	chennai
5.	40.0000	2500.000	Silver	chennai
6.	35.0000	1300.000	Gold	mumbai
7.	35.42857	1600.000	Silver	Kolkata

```
# Print final cleaned dataset
```

```
print (final_data).
```

Result:

Thus, the program successfully demonstrates importing data from an Excel workbook and a relational database, integrating multiple sources into a single dataset, exploring the data structure and identifying missing values and duplicates, separating numerical and categorical data, cleaning missing values and duplicates, converting categorical variables into factors, and producing a clean dataset ready for accurate business analysis.